

UI/UX Design

Document 03

Status	Draft v1
Stack	Gradio Blocks mounted inside FastAPI (v1). Custom React front end only if Gradio blocks a P0 requirement
Related	01 PRD , 04 App Flow

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1 Design principles

1. **Verify, don't type.** Every text box opens pre-filled (aligned transcript or model hypothesis). The fastest action is “approve”.
2. **Keyboard first.** A reviewer should process a segment without touching the mouse.
3. **The model suggests, the human decides.** Model output is always visually distinct (a diff panel) and is never silently written into the editable text.
4. **One primary action per screen,** shown with the primary colour. Destructive actions are secondary and need confirmation.
5. **State is always visible.** The segment's status, whether there are unsaved changes, job progress and the active model are shown on screen.

2 Visual language

This keeps the earth-tone theme already chosen in review/app.py and formalises it as tokens.

Token	Light	Dark	Use
--bg	#f4eee5	#211b17	page
--surface	#fffaf3	#30261f	cards, inputs
--text	#2d241d	#f4eee5	body
--muted	#675443	#d5bda5	secondary text, timestamps
--border	#b99b7a	#806149	dividers
--accent	#8a5a3b	#b77b51	primary buttons, focus ring

Token	Light	Dark	Use
--accent-hover	#6f432d	#d19669	
--ok	#4f7a3a	#8fbf6f	approved
--warn	#b07d1a	#e0b04f	flagged, low confidence
--danger	#9c3b2e	#e07a6a	rejected, errors
--lang-en	#2f5d8a underline	#7fb0e0	language span tint
--lang-sw	#4f7a3a underline	#8fbf6f	
--lang-mixed	#8a3b7a underline	#d08ac0	

Typography: Times New Roman (the project's choice) for UI text at 12 pt minimum. The transcript editor uses **14 pt** with a line height of 1.6, because reviewers read it for hours. Timestamps and IDs use a monospace font at 11 pt. Colour is never the only signal: flags also show an icon and a label.

3 Information architecture

```

Top bar: [Transcribe] [Corpus Studio] [Models & Evaluation]      active model: whisper-small-
↳ kct-v1 ●
  |
  |-- Transcribe
  |   |-- New transcription (upload)
  |   |-- Jobs list
  |   `-- Transcript editor
  |-- Corpus Studio
  |   |-- Recordings list
  |   |-- Import recording (wizard)
  |   |-- Recording overview (alignment report)
  |   |-- Review (segment reviewer)  <- most-used screen
  |   `-- Datasets (build / export snapshots)
  `-- Models & Evaluation
      |-- Model registry
      |-- Evaluation report
      `-- Compare two models

```

4 Screens

4.1 Transcribe: New transcription

```

+-----+
| Drop audio or video here, or [Browse] |
| WAV MP3 M4A MP4 OGG FLAC · up to 3 h |
+-----+
| Title [_____] Domain [Interview v] |
| Model [whisper-small-kct-v1 (production) v] |
| [ ] Allow corrections to be added to training data (consent) |
|                                     [ Start transcription ] |
+-----+

```

- The consent checkbox is off by default and maps to consent_scope.
- After submitting, the user lands on the Jobs list with a progress bar: “Transcribing 12:40 / 58:10 · ETA 18 min”.

4.2 Transcribe: Transcript editor

```

+-----+
| Sermon_2026-09-14.mp3 · 58:10 · whisper-small-kct-v1 [Export v]|
+-----+
| ▶ || 00:12:41 ————— 00:58:10 1.0x v |
+-----+
| 00:12:31 S? | Leo tunataka ku-focus kwa neno la Mungu, |
| 00:12:38 S? | and I want you to open your Bibles |
|>00:12:41 | pale Mathayo sura ya tano. [edited] |
| 00:12:49 | Wale wa nyuma mnasikia? low confidence |
+-----+
| Enter: play segment · Ctrl+S: save · Alt+Up/Down: prev/next |
| Ctrl+Shift+S: split at cursor · Ctrl+M: merge with next |
+-----+

```

- Clicking a segment seeks the player there, and the current segment is highlighted during playback.
- Low-confidence words (word probability < 0.5) get a dotted - - warn underline, with the probability shown in a tooltip.
- Edited segments show an [edited] chip. Edits autosave after 1.5 s idle.
- The Export menu offers TXT, SRT, VTT, DOCX, JSON, plus “Promote edits to corpus” when consent allows.

4.3 Corpus Studio: Import recording (3-step wizard)

1. **Files:** audio (required), transcript DOCX/PDF (optional).
2. **Metadata:** title, domain, recorded date, region (free text, optional), consent scope (required: *training / eval only / transcription only*), and the language-hint convention for the DOCX (“*italic = Kiswahili*” ✓).
3. **Confirm:** a parsed preview showing the first 10 turns with speaker labels, the count of turns, the count of [unclear] markers, and the audio duration. Pressing **Import & align** enqueues the job.

4.4 Corpus Studio: Recording overview

```

KSM09 · interview · 1:12:04 · consent: training
Alignment: done · mean confidence 0.81 · 612 segments ██████████
[ ] 402 approved · 31 flagged · 179 to review
Low-confidence turns (likely non-verbatim): p0142 0.31 | p0388 0.28 | ...
Confidence timeline: ██████████ (click to jump)
                        [ Start reviewing ▶ ]

```

The confidence timeline shows problem regions (non-verbatim passages, music, crosstalk) at a glance.

4.5 Corpus Studio: Segment reviewer (primary screen)

```

+-----+
| KSM09 · segment 143/612 · priority 0.71 · status: candidate |
| Queue: [Priority v] Filter: [All flags v] unsaved ● |
+-----+
| waveform ██████████ 00:14:02.35 → 00:14:13.90 |
| ◀◀[ -250ms] ◀[ -50ms] start [+50ms ▶] end ◀[ -50ms] [+50ms ▶]|
| ▶ Play (Tab) ♻️ Replay last 3s (Shift+Tab) 0.75x / 1x |
+-----+
| Transcript (editable) 14 pt |
| |
| | Sasa mimi nilikuwa nimeapply hiyo job lakini they never |
| | called back, so nikaamua tu kuanza biashara yangu. |
| |
+-----+

```

```

| |
| Source: aligned transcript -p0141p0142 · word conf 0.42 "nimeapply" |
+-----+
| Model hypothesis (whisper-small-kct-v1) [Run model (R)] |
| Sasa mimi nilikuwa nimeapply hiyo job lakini they never |
| called back so ~nikaamua~ **nikamua** tu kuanza biashara yangu |
| WER vs transcript: 7.1% [Use hypothesis text (Ctrl+H)] |
+-----+
| Flags: [U]nclear [O]verlap [N]oisy [M]usic [V]non-verbatim |
| Language spans (select text, then E / S / X mixed / L Luo) |
| Speaker: interviewer ▾ |
+-----+
| [Reject (Ctrl+Del)] [Save draft (Ctrl+S)] [Approve & next ↵] |
| History ▾ r3 you 14:02 · r2 aligner · r1 aligner |
+-----+

```

Behaviour:

- **Approve & next** (Ctrl+Enter) saves a revision with status=approved and loads the next item in the queue. Tab plays the segment. The segment auto-plays on load (a setting, on by default).
- Changing a boundary re-renders the slice and replays the last 3 s (for the end boundary) or the first 3 s (for the start boundary).
- **Use hypothesis text** is the only way hypothesis text enters the editor. It is explicit, and it creates a revision with source=model_hypothesis.
- Flags are toggles, with the key shown in brackets. Setting *Unclear* or *Non-verbatim* shows an inline note: “Excluded from training until resolved.”
- Pressing Approve on a segment that has an active exclusion flag shows a confirmation: “Approve for eval only?”
- The History drawer lists revisions and offers “Restore r2” (which creates r4 as a copy of r2, so nothing is ever deleted).
- A 409 conflict shows a banner: “This segment changed in another tab. Reload / Keep mine as new revision.”

Keyboard map:

Key	Action
Tab / Shift+Tab	Play segment / replay last 3 s
Ctrl+Enter	Approve & next
Ctrl+S	Save draft
Ctrl+Del	Reject
Alt+↑ / Alt+↓	Previous / next in queue
[]	Start -50 ms / +50 ms
{ }	End -50 ms / +50 ms
Alt+U/O/N/M/V	Toggle flags
Alt+E/S/X/L	Tag selected text as en / sw / mixed / Luo
R	Run the model on the segment
Ctrl+H	Copy hypothesis into the editor

The shortcuts are bound with a small injected JS snippet (`gr.Blocks(js=...)`), because Gradio has no native global hotkeys.

4.6 Corpus Studio: Datasets

- **Filter builder:** recordings (multi-select), domains, consent = training (locked), status = approved, exclude flags (default: unclear, non_verbatim, overlap), speaker roles (default: all), min/max duration.
- **Split:** “Use frozen test set v1” is locked. Dev % and seed are settable. A preview table shows each split's hours, segments, recordings and code-switch density, with a leakage check ✓.
- **Build snapshot** creates kct-ds-v3 (frozen). **Export for Kaggle** produces a zip of the folder plus a README with the dataset hash.

4.7 Models & Evaluation

- **Registry table:** name, base model, dataset snapshot, dev WER, test WER, status (candidate/production/retired), and actions (Evaluate, Promote, Convert to CPU format).
- **Evaluation report:** metric cards (WER, CER, switch-point, hallucination/min, RTF), a breakdown table by domain × flag, and the worst 50 segments with audio, reference/hypothesis diff, and a “Send to review” action.
- **Compare:** two models on the same split, showing the per-metric delta, segments where A beats B and vice versa, and error categories (substitution en→sw and sw→en, deletions at switch points).

5 States and feedback

State	Treatment
Job queued/running	Progress bar with processed/total seconds and ETA; the page can be closed
Job failed	Red card with the error summary, a “Retry” action and the log link
No production model	Transcribe is disabled with the notice: “Register a model in Models & Evaluation”; Review still works without “Run model”
Empty review queue	“All segments reviewed. Build a dataset snapshot →”
Unsaved changes	A dot in the header, and a browser beforeunload warning
Offline aligner model missing	A setup checklist on first run (ffmpeg, aligner weights, VAD weights)

6 Accessibility

- Every action is keyboard-accessible, and the focus ring is a 2 px - -accent outline.
- Contrast is at least 4.5:1 for text in both themes (the tokens above meet this for body text on surface).
- Audio controls have labels; screen-reader text is set on the flag toggles.
- Playback speed runs from 0.5× to 1.5× to help with fast or accented speech.

7 Gradio feasibility notes

Need	Approach
Waveform + boundary nudging	gr.Audio waveform for playback; the boundaries are numeric fields with nudge buttons (slices are re-rendered server-side). No drag handles in v1
Diff rendering	gr.HTML with a server-side word diff (difflib on normalised tokens)
Language span tagging	gr.HighlightedText for display; tagging through selection + hotkey handled by JS, which posts (start, end, lang)
Global hotkeys	gr.Blocks(js=...) keydown listener that clicks hidden buttons
Multi-page	gr.Tabs for the three surfaces, with FastAPI serving the downloads

If boundary dragging or span tagging proves too clumsy in Gradio after M1, those two widgets move to a small custom component (`gradio_cc`) rather than a full front-end rewrite.